Word Embeding techniques

- we have studied counter frequency like tfidf, bag of words, 1 hot encode these are not models but counting techniques or simple programmings.
- we can also use DL models like
- a book for GenAI is suggested by sir “Applied Gen AI for Beginners”

Word Embedings In NLP

- previous we study in which there was higher dimensions we need to do in lower dimensions
- it is an approach to represent words and documents
- sparse matrix is mostly 0 few 1s
- reference geeks for geeks word embedings in nlp
- word2vec and gloVe are mostly used in NLPs

- The Word2Vec also itself creates feature representation for the features present in our data.

Feature	Boy	Girl	King		Apple	Mango
Gender	0.01	0.02	0.95	0.96	-0.02	0
Royalty	0.01	0.01	0.95	0.96	-0.02	0
Age	0.03	0.02	0.07	0.08	0	0
Food	0	0	0	0	0.94	0.96

- next class task pg no. 334, deep learning book chapter 11 deep learning for text latest ver deep learning book

```python
import numpy as np
path_to_glove_file = "glove.6B.100d.txt"

embeddings_index = {}
with open(path_to_glove_file) as f:
    for line in f:
        word, coefs = line.split(maxsplit=1)
        coefs = np.fromstring(coefs, "f", sep=" ")
        embeddings_index[word] = coefs

print(f"Found {len(embeddings_index)} word vectors.")
```

Next, let's build an embedding matrix that you can load into an `Embedding` layer. It must be a matrix of shape `(max_words, embedding_dim)`, where each entry *i* contains the `embedding_dim`-dimensional vector for the word of index *i* in the reference word index (built during tokenization).

- complete all the codes till embeddings
- download the given vocabulary files